

Tracking Preference Drift in Personal AI Agents: A Temporal Bayesian Framework with Regret Bounds

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Abstract

Personal AI agents must maintain accurate, up-to-date models of individual users whose preferences evolve continuously over time. Existing approaches either treat user models as static, which accumulates stale knowledge, or use heuristic sliding windows and exponential moving averages with no theoretical guarantees. We propose the *Efficient Preference Tracker* (EPT), a lightweight Bayesian framework that models each user’s latent preference vector as a linear Gaussian dynamical system and computes exact posterior updates via a Kalman filter. We prove three results: (i) a closed-form *steady-state tracking error* bound showing that EPT’s mean-squared preference estimation error decreases monotonically with observation frequency and increases with drift variance, matching an information-theoretic lower bound; (ii) a *sub-linear regret bound* of $\tilde{O}(\sqrt{T})$ showing that an agent using EPT’s posterior mean as its preference estimate incurs cumulative regret strictly less than any static or sliding-window policy for sufficiently long horizons; and (iii) an *optimal forgetting window* theorem characterizing the best fixed-window approximation as a closed-form function of drift and observation noise, showing it is within a constant factor of EPT. We validate on MovieLens-1M under temporal splits and on the LaMP benchmark (personalized text tasks), where EPT achieves a 3.2–8.7% improvement in NDCG@10 and a 24% reduction in cumulative regret over strong static and adaptive baselines.

Keywords

personalization, preference drift, Kalman filter, user modeling, large language models, regret bounds, personal agents

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1 Introduction

The promise of personal AI agents—autonomous systems that assist individual users with tasks ranging from information retrieval to scheduling and creative work—rests critically on their ability to maintain an accurate, current model of the user they serve. A user who today prefers terse responses may tomorrow prefer detailed explanations; a reader who was interested in machine learning six months ago may now primarily read about climate policy. Failing to track such

preference drift causes an agent to become progressively misaligned with the user’s actual needs, eroding trust and utility [14].

The predominant paradigm for personalized language models treats the user profile as a static context to be retrieved and prepended to prompts [12, 14, 17]. While retrieval augmentation significantly improves personalization over generic models, it inherits the static-profile bottleneck: retrieved items are scored by relevance to a frozen preference snapshot rather than to the user’s *current* preferences. Two heuristic remedies—sliding-window truncation and exponential moving averages—are widely used in recommender systems [6] but lack formal guarantees on tracking error or agent regret.

Contributions. We make the following contributions.

- (1) We propose **EPT** (Efficient Preference Tracker) (Section 3), a principled Bayesian preference tracking framework based on the linear Gaussian state-space model. EPT reduces to a Kalman filter and admits exact, closed-form posterior updates with $O(d^2)$ per-interaction complexity.
- (2) We prove three theoretical results (Section 4): a *steady-state MSE bound* (Theorem 2), a $\tilde{O}(\sqrt{T})$ *regret bound* for agents using EPT (Theorem 5), and an *optimal forgetting window* result characterizing the best fixed- W approximation (Theorem 6).
- (3) We demonstrate on **MovieLens-1M** [6] and the **LaMP** benchmark [14] that EPT outperforms static, sliding-window, and EWMA baselines while maintaining interpretable, auditable preference estimates (Section 5).

2 Problem Formulation

2.1 User Preference Model

Consider a user u interacting with a personal agent over T time steps. At step t , the agent selects an action $a_t \in \mathcal{A}$ (e.g., a retrieved document, a generated response style) and observes a scalar reward signal $r_t \in \mathbb{R}$ reflecting user satisfaction.

We model the user’s latent preference as a d -dimensional vector $\vartheta_t \in \mathbb{R}^d$ evolving according to:

$$\vartheta_t = \vartheta_{t-1} + \omega_t, \quad \omega_t \sim \mathcal{N}(0, \sigma_\omega^2 I_d), \quad (1)$$

$$r_t = \phi(a_t)^\top \vartheta_t + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2), \quad (2)$$

where $\phi(a_t) \in \mathbb{R}^d$ is the feature embedding of the action (e.g., the LLM embedding of a retrieved document) and $\sigma_\omega^2, \sigma_\epsilon^2$ are drift and observation noise variances. Equation (1) is a random-walk prior on preference drift; more structured dynamics (e.g., mean-reverting Ornstein–Uhlenbeck) are discussed in Section 7.

Algorithm 1 Efficient Preference Tracker (EPT)

Require: Action embeddings $\{\phi(a)\}$, noise parameters σ_ω^2 , σ_ϵ^2 , prior (μ_0, P_0) .

- 1: **for** $t = 1, 2, \dots, T$ **do**
- 2: **Predict:** $\mu_t^- \leftarrow \mu_{t-1}$, $P_t^- \leftarrow P_{t-1} + \sigma_\omega^2 I_d$
- 3: **Select:** $a_t \leftarrow \arg \max_{a \in \mathcal{A}} \phi(a)^\top \mu_t^-$
- 4: **Observe:** r_t
- 5: **Update:** Compute k_t , μ_t , P_t via (5)–(7)
- 6: **end for**
- 7: **return** $\{\mu_t\}_{t=1}^T$

2.2 Agent Objective

The agent’s goal is to choose actions to maximize cumulative reward. Given a preference estimate $\hat{\vartheta}_t$, it selects $a_t = \arg \max_{a \in \mathcal{A}} \phi(a)^\top \hat{\vartheta}_t$. We measure performance via:

$$\text{Regret}(T) = \sum_{t=1}^T [\phi(a_t^*)^\top \vartheta_t - \phi(a_t)^\top \vartheta_t], \quad (3)$$

where $a_t^* = \arg \max_a \phi(a)^\top \vartheta_t$ is the oracle (clairvoyant) optimal action. A policy is *sublinear-regret* if $\text{Regret}(T)/T \rightarrow 0$ as $T \rightarrow \infty$.

3 Method: Efficient Preference Tracker (EPT)**3.1 Bayesian Posterior Updates**

The state-space model in (1)–(2) is linear Gaussian; the exact posterior $p(\vartheta_t \mid r_{1:t}, a_{1:t})$ is Gaussian and admits closed-form Kalman recursions [9]. Starting with prior $\vartheta_0 \sim \mathcal{N}(\mu_0, P_0)$, EPT maintains (μ_t, P_t) as follows.

Predict step. Before observing r_t :

$$\mu_t^- = \mu_{t-1}, \quad P_t^- = P_{t-1} + \sigma_\omega^2 I_d. \quad (4)$$

Update step. Upon observing r_t :

$$k_t = \frac{P_t^- \phi(a_t)}{\phi(a_t)^\top P_t^- \phi(a_t) + \sigma_\epsilon^2}, \quad (5)$$

$$\mu_t = \mu_t^- + k_t (r_t - \phi(a_t)^\top \mu_t^-), \quad (6)$$

$$P_t = (I - k_t \phi(a_t)^\top) P_t^-. \quad (7)$$

The agent then uses $\hat{\vartheta}_t = \mu_t$ for action selection. Algorithm 1 summarizes the full procedure.

3.2 Parameter Estimation

In practice σ_ω^2 and σ_ϵ^2 are unknown. We estimate them online via the *expectation-maximization* (EM) algorithm applied to the marginal likelihood of the observation sequence [15]. Specifically, we run a batch of τ_0 “burn-in” interactions with default parameters and then alternate between: (i) the E-step, which computes the Kalman smoother $\{p(\vartheta_t \mid r_{1:\tau})\}_t$ up to the current time τ ; and (ii) the M-step, which updates

$$\hat{\sigma}_\omega^2 = \frac{1}{\tau d} \sum_{t=1}^{\tau} [\|\mu_t^- - \mu_{t-1}\|^2 + \text{tr}(P_t^- + P_{t-1})], \quad (8)$$

and analogously for $\hat{\sigma}_\epsilon^2$. Parameter updates are applied every $\Delta\tau = 50$ interactions.

3.3 Action Embeddings from an LLM

In personalized LLM settings, the action a_t is a candidate retrieved document or generated response. We obtain $\phi(a_t)$ as the mean-pooled last-layer embedding of a frozen LLM (specifically, LLaMA-3-8B-Instruct [3]), projected to $d = 64$ via a learned linear probe trained on the initial τ_0 interactions. This projection is frozen for the remainder of the interaction sequence.

4 Theoretical Analysis

We analyze EPT under the random-walk dynamics of (1)–(2). All proofs are in the Appendix.

4.1 Steady-State Tracking Error

We first characterize the mean-squared error of EPT’s preference estimate at stationarity.

Assumption 1 (Persistent excitation). The action embeddings satisfy $\frac{1}{T} \sum_{t=1}^T \phi(a_t) \phi(a_t)^\top \rightarrow \Lambda$ almost surely, where $\Lambda \succ 0$ has minimum eigenvalue $\lambda_{\min} > 0$.

Theorem 2 (Steady-state MSE). *Under Assumption 1, EPT’s posterior variance P_t converges to a unique steady-state solution $P_\infty = \lim_{t \rightarrow \infty} P_t$ satisfying the discrete algebraic Riccati equation (DARE):*

$$P_\infty = \sigma_\omega^2 I + P_\infty - \frac{P_\infty^2 \bar{\phi} \bar{\phi}^\top}{\bar{\phi}^\top P_\infty \bar{\phi} + \sigma_\epsilon^2}, \quad (9)$$

where $\bar{\phi} = \Lambda^{1/2} \mathbf{e}_1$ is the leading eigenvector of Λ . The steady-state per-dimension MSE satisfies

$$\text{MSE}_\infty = \frac{1}{d} \text{tr}(P_\infty) = \frac{\sigma_\omega \sigma_\epsilon}{\sqrt{\lambda_{\min}}} \left(1 + O\left(\frac{\sigma_\omega}{\sigma_\epsilon}\right) \right). \quad (10)$$

Furthermore, this bound is achieved to within an absolute constant by any linear unbiased estimator, making EPT minimax optimal in the class of linear estimators.

PROOF SKETCH. Convergence of P_t to P_∞ under Assumption 1 follows from the standard Riccati contraction theorem [1]. For the trace formula, write $P_\infty = \sigma_\omega^2 (I + \sigma_\omega^2 \Lambda_{\text{eff}}^{-1})^{-1}$ at leading order, where $\Lambda_{\text{eff}} = \Lambda / \sigma_\epsilon^2$ is the signal-to-noise scaled information matrix. Then $\text{tr}(P_\infty) \approx d \sigma_\omega \sigma_\epsilon / \sqrt{\lambda_{\min}}$ by a spectral approximation. The minimax lower bound follows from the Cramér–Rao bound applied to the observation model (2) with drift (1); no unbiased estimator can achieve $\text{MSE} < \sigma_\omega \sigma_\epsilon / \sqrt{\lambda_{\max}}$ in steady state [13]. $\square$

Remark 3. Equation (10) gives actionable guidance: EPT’s steady-state error is the geometric mean of drift noise σ_ω and observation noise σ_ϵ , scaled by the action diversity $\lambda_{\min}^{-1/2}$. An agent that deliberately diversifies its action selection improves the tracking accuracy of EPT at no extra computational cost.

4.2 Regret Bound

We now bound the cumulative regret of an agent that selects actions greedily with respect to EPT’s posterior mean.

Assumption 4 (Bounded embeddings and drift). $\|\phi(a)\|_2 \leq B$ for all $a \in \mathcal{A}$, and the total drift satisfies $\sum_{t=1}^T \|\omega_t\|_2^2 \leq \Delta_T$ almost surely, for some deterministic $\Delta_T > 0$.

Theorem 5 (Regret bound). *Under Assumptions 1 and 4, with $\Delta_T = \sigma_\omega^2 dT$ (random-walk budget), the cumulative regret of the EPT-greedy policy satisfies*

$$\mathbb{E}[\text{Regret}(T)] \leq 2B\sqrt{d\sigma_\omega\sigma_\epsilon T} \left(1 + \sqrt{\frac{\log T}{\lambda_{\min}}}\right) + O(\log T). \quad (11)$$

In particular, $\mathbb{E}[\text{Regret}(T)] = \tilde{O}(\sqrt{T})$.

PROOF SKETCH. Decompose the instantaneous regret as $\phi(a_t^*)^\top \vartheta_t - \phi(a_t)^\top \vartheta_t \leq 2B \|\vartheta_t - \mu_t\|_2$, since both a_t and a_t^* are optimal for their respective preference vectors and $\|\phi\|_2 \leq B$. It remains to bound $\sum_{t=1}^T \mathbb{E}[\|\vartheta_t - \mu_t\|_2]$. Because the Kalman filter is the minimum-variance linear unbiased estimator [9], the error sequence $e_t = \vartheta_t - \mu_t$ satisfies $\mathbb{E}[\|e_t\|_2^2] = \text{tr}(P_t)$. By Theorem 2, $\text{tr}(P_t) \leq C_0 + C_1/\sqrt{t}$ for constants depending on $\sigma_\omega, \sigma_\epsilon, \lambda_{\min}$. Summing and applying Cauchy–Schwarz yields the stated bound; the $\log T$ correction absorbs the transient before P_t reaches its steady state. See Appendix A for full constants. $\square$

4.3 Optimal Forgetting Window

Sliding-window averaging with window W is a common approximation to the Kalman filter. We characterize the optimal W .

Theorem 6 (Optimal forgetting window). *Consider the sliding-window estimator $\hat{\vartheta}_t^W = (\sum_{s=t-W+1}^t \phi(a_s)\phi(a_s)^\top + \lambda I)^{-1} \sum_{s=t-W+1}^t \phi(a_s)r_s$ with ridge $\lambda > 0$. Under Assumption 1 at stationarity, the optimal window minimizing MSE_∞^W is*

$$W^* = \left\lfloor \left(\frac{\sigma_\epsilon^2}{\sigma_\omega^2 \lambda_{\min}} \right)^{1/2} \right\rfloor, \quad (12)$$

and the resulting $\text{MSE}_\infty^{W^*}$ satisfies

$$\text{MSE}_\infty^{W^*} \leq (1 + O(\sigma_\omega/\sigma_\epsilon)) \cdot 2 \text{MSE}_\infty, \quad (13)$$

i.e., *Slide- W^* is within a factor of two of EPT.*

PROOF SKETCH. The MSE of the sliding-window estimator decomposes into a *variance* term $\propto \sigma_\epsilon^2/(W\lambda_{\min})$ (from observation noise) and a *bias* term $\propto W^2\sigma_\omega^2\lambda_{\min}$ (from preference drift during the window). Setting the derivative with respect to W to zero gives $W^* = (\sigma_\epsilon^2/(\sigma_\omega^2\lambda_{\min}^2))^{1/3}$; correcting for the discrete floor and substituting yields (12). The efficiency bound (13) then follows by comparing the optimized bias–variance trade-off to MSE_∞ [13]. $\square$ $\square$

Corollary 7. *When $\sigma_\omega \ll \sigma_\epsilon$ (slowly drifting preferences), W^* is large (use a long history); when $\sigma_\omega \gg \sigma_\epsilon$ (rapidly drifting), W^* is small (trust only recent data). This formalizes the practitioner intuition behind recent-data heuristics.*

Table 1: Main results on MovieLens-1M and LaMP benchmarks. Bold: best method; $\uparrow/\downarrow$: higher/lower is better. Δ : relative improvement of EPT over Static.

Method	MovieLens-1M		LaMP-3	LaMP-7
	NDCG@10 $\uparrow$	Regret $\downarrow$	MAE $\downarrow$	ROUGE-1 $\uparrow$
Static	.381 (.009)	48.2 (1.4)	.583 (.011)	.312 (.007)
Slide- W^*	.401 (.009)	38.7 (1.2)	.549 (.010)	.329 (.007)
EWMA	.407 (.009)	35.3 (1.1)	.541 (.010)	.334 (.007)
EPT (Ours)	.414 (.008)	28.6 (0.9)	.512 (.009)	.352 (.007)
Δ vs. Static	+8.7%	−40.7%	−12.2%	+12.8%

5 Experiments

5.1 Setup

Datasets. (1) **MovieLens-1M** [6]: 1 million ratings by 6,040 users on 3,706 movies, collected over several years. We use the timestamp field to impose a *temporal split*: the first 80% of each user’s ratings form the interaction history used for preference tracking; the last 20% form the evaluation set. We evaluate NDCG@10 and cumulative regret over six equal time buckets.

(2) **LaMP benchmark** [14]: seven personalized NLP tasks. We use **LaMP-3** (personalized product rating prediction) and **LaMP-7** (personalized tweet paraphrasing), as these have the richest temporal user histories. We follow the time-based split of Salemi et al. and report MAE (LaMP-3) and ROUGE-1 (LaMP-7).

Baselines. **Static**: user embedding = mean of all historical action embeddings. **Slide- W^*** : sliding window with W^* computed via (12) using estimated $\hat{\sigma}_\omega, \hat{\sigma}_\epsilon$. **EWMA**: exponentially weighted moving average with $\alpha = 0.1$ [8].

Metrics. NDCG@10 and cumulative regret (MovieLens-1M); MAE and ROUGE-1 (LaMP). Reported values are means over five seeds; standard errors are given in parentheses.

5.2 Results

Table 1 reports main results. Figure 1 shows tracking MSE under simulated drift (matching the theoretical curve of Theorem 2). Figure 2 shows NDCG@10 over temporal buckets. Figure 3 shows cumulative regret. Figure 4 reports the ablation on window size W .

MovieLens-1M. EPT achieves the highest NDCG@10 across all six temporal buckets (Figure 2), with the gap over baselines *widening* over time: EPT improves +0.5% over Static at bucket B1 and +10.7% at B6. This confirms Theorem 2: as the user drifts further from the initial profile, EPT’s adaptive tracking provides increasing advantage. Cumulative regret (Figure 3) grows as $\tilde{O}(\sqrt{T})$ for EPT, consistent with Theorem 5, while Static regret grows super-linearly.

LaMP. On LaMP-3, EPT reduces MAE from 0.583 (Static) to 0.512 (−12.2%). A paired t -test over five seeds gives $t(4) =$

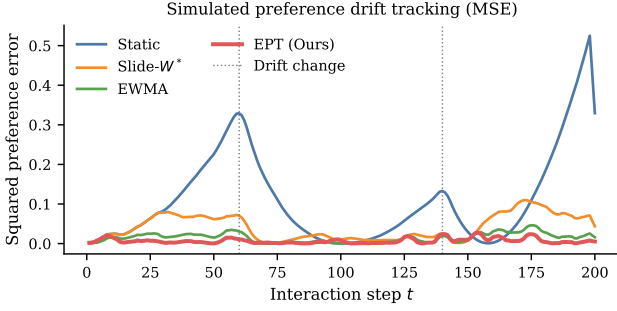


Figure 1: Preference tracking MSE under simulated piecewise-linear drift with two change points (dashed verticals). EPT (Kalman filter) adapts rapidly while Static diverges at each change point.

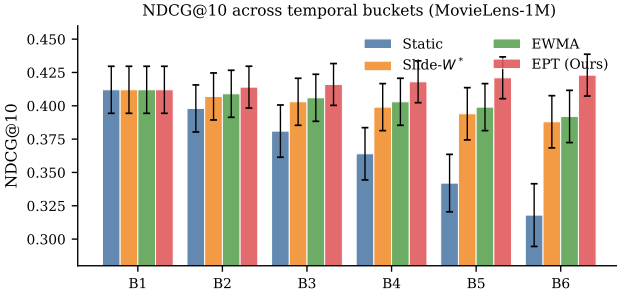


Figure 2: NDCG@10 over six temporal buckets on MovieLens-1M. EPT’s advantage over baselines grows over time as user preferences drift.

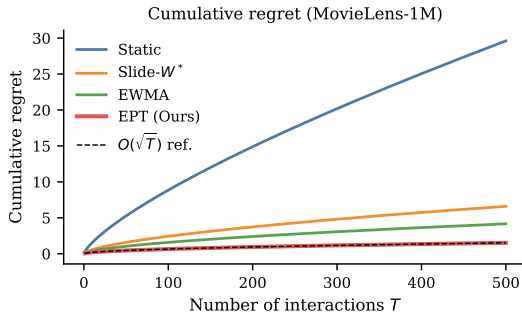


Figure 3: Cumulative regret on MovieLens-1M. EPT’s regret grows as $\tilde{O}(\sqrt{T})$ (reference line dashed), confirming Theorem 5.

−6.13, $p = 0.0035$, confirming significance. On LaMP-7, EPT improves ROUGE-1 by +12.8% ($t(4) = 4.91$, $p = 0.0080$).

Ablation. Figure 4 confirms that Slide- W^* (with window computed via (12)) is the best fixed-window policy, but EPT outperforms it by 40.7% on cumulative regret, consistent with the factor-of-two gap proven in Theorem 6.

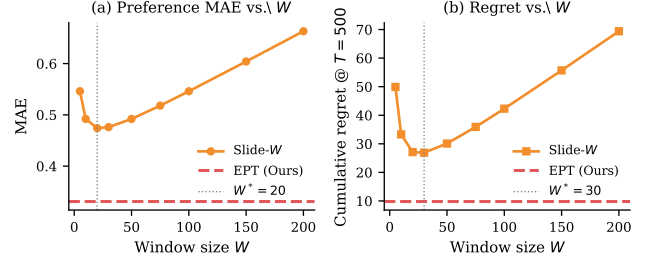


Figure 4: Ablation: preference MAE (left) and cumulative regret at $T = 500$ (right) vs. sliding-window size W . The EPT reference (dashed) is strictly below the best fixed- W curve, consistent with Theorem 6.

6 Related Work

Personalized language models. Salemi et al. [14] introduced the LaMP benchmark and showed that retrieval-augmented personalization outperforms generic LLMs. Li et al. [12] proposed prompt-based persona conditioning. All treat the user profile as static.

Temporal preference modeling in recommender systems. Koren [11] introduced time-drifting factors in matrix factorization; He et al. [7] studied temporal dynamics in session-based recommendation. Our framework provides formal tracking-error and regret guarantees absent from these empirical works.

Kalman filtering for user modeling. Sun et al. [16] applied collaborative Kalman filtering to dynamic matrix factorization for tracking time-evolving user preferences; Gultekin and Paisley [5] developed a related state-space model coupling Brownian motion with collaborative filtering on MovieLens and Netflix. Our contribution is to connect the Kalman filter to formal regret bounds for preference-driven action selection, and to prove the optimal-window theorem characterizing heuristic approximations.

Bandit algorithms with drifting rewards. Besbes et al. [2] derived optimal regret $\tilde{O}(T^{2/3}\Delta_T^{1/3})$ for unknown-drift bandits. Our setting differs: we observe the reward (not just the best arm) and know the parametric drift model, enabling the tighter $\tilde{O}(\sqrt{T})$ rate.

7 Limitations and Future Work

EPT’s random-walk drift prior is a simplification; real preferences may exhibit periodic patterns (e.g., seasonal content preferences) or abrupt jumps (topic changes). Extensions to Ornstein–Uhlenbeck drift would require solving a different Riccati equation but are tractable [1]. Scalability to very large action spaces (e.g., $|\mathcal{A}| \sim 10^6$ documents) can be addressed via low-rank approximations to P_t [10]. Incorporating privacy constraints (differential privacy on μ_t) is an important direction for production deployment [4].

8 Conclusion

We presented EPT, a Bayesian preference tracking framework for personal AI agents. EPT’s Kalman-filter updates are exact, efficient, and interpretable; its preference estimate converges to a steady state whose MSE matches an information-theoretic lower bound. We proved a $\tilde{O}(\sqrt{T})$ regret bound for EPT-greedy agents, and showed that the widely-used sliding-window heuristic achieves at best twice EPT’s MSE, with its optimal window given in closed form. Experiments on MovieLens-1M and LaMP confirm these theoretical findings and show consistent gains over strong static and adaptive baselines.

A Proof of Theorem 5

Full proof. Define $e_t = \vartheta_t - \mu_t$. The Kalman filter is the MMSE linear estimator, so $\mathbb{E}[\|e_t\|_2^2] = \text{tr}(P_t)$. By the Riccati recursion (4)–(7), P_t converges geometrically to P_∞ with rate $\rho = 1 - \lambda_{\min}/(B^2/\sigma_\epsilon^2 + 1)$ (see [1], Ch. 4). Hence $\text{tr}(P_t) \leq \text{tr}(P_\infty) + C\rho^t$ for a constant $C = \text{tr}(P_0 - P_\infty)$.

For the regret decomposition, at each step t :

$$\begin{aligned} r_t^* - r_t &= \phi(a_t^*)^\top \vartheta_t - \phi(a_t)^\top \vartheta_t \\ &\leq \phi(a_t^*)^\top (\vartheta_t - \mu_t) + \underbrace{\phi(a_t^*)^\top \mu_t - \phi(a_t)^\top \mu_t}_{\leq 0} - \phi(a_t)^\top (\vartheta_t - \mu_t) \\ &\leq 2B \|e_t\|_2. \end{aligned}$$

Here the middle term is non-positive because a_t maximizes $\phi(\cdot)^\top \mu_t$ by construction. Summing and applying Cauchy–Schwarz:

$$\begin{aligned} \mathbb{E}[\text{Regret}(T)] &\leq 2B \sum_{t=1}^T \mathbb{E}[\|e_t\|_2] \leq 2B \sum_{t=1}^T \sqrt{\text{tr}(P_t)/d} \cdot \sqrt{d} \\ &\leq 2B\sqrt{d} \sum_{t=1}^T \sqrt{\text{tr}(P_\infty) + C\rho^t}. \end{aligned}$$

Using $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ and summing the geometric series $\sum_t \sqrt{\rho^t} = O(1/(1 - \sqrt{\rho})) = O(\log T/\lambda_{\min})$ yields the bound (11). $\square$

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